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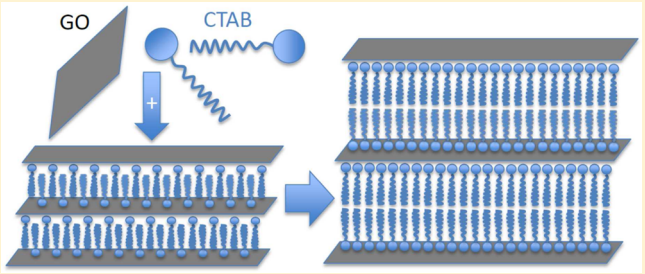
Structure and Interaction of Graphene Oxide–Cetyltrimethylammonium Bromide Complexation

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Supporting Information

ABSTRACT: Complexation of graphene oxide (GO) and cetyltrimethylammonium bromide (CTAB), a model amphiphile, was quantitatively studied to elucidate the governing forces and resulting structures of their self-assembly. We systematically varied pertinent self-assembly parameters, including GO size, mixing ratio, salt, and osmotic stress, and quantified the nanoscale structures of GO:CTAB using X-ray diffraction. Multilamellar stacking of GO and CTAB layers persists in all GO:CTAB assemblies exhibiting structural order. While driven by hydrophobic interactions between the hydrocarbon tails, the CTAB layer displays a series of structural transitions from a bilayer with in-plane ordering to a more fluidic bilayer and to an interdigitated monolayer. Interaction between GO and CTAB layers shows domination by electrostatics, and osmotic stress studies indicate the absence of continuous hydration layers in GO:CTAB assemblies. Adding salt increases the interlayer spacing by intercalating between GO and CTAB layers, and this ion layer can be expelled by high osmotic stress. These studies provide quantitative knowledge of the structure and interaction of GO:CTAB complexation, which lays foundation for precise control and rational design of their nanoscale self-assemblies.



INTRODUCTION

Materials of complex structures and multiple functionalities are an increasingly important part of future material needs. One of the promising areas is carbon-based nanomaterials (CNM) and their nanoscale assembly.[1,2] CNMs are a family of carbon allotropes with extremely diverse structures and functional properties, ranging from amorphous carbon, graphite, and buckyball to the more recent carbon nanotube and graphene. To realize the potential of CNMs, major efforts are being invested on many fronts, including functional enhancements of CNMs through synthesis, sorting, chemical modifications, and biomolecular hybridization as well as molecular self-assembly of CNMs into functional nanostructures and devices.[1,2]

Self-assembly is a key concept for organizing CNMs via "bottom up" methods to form structures beyond the limit of conventional "top-down" methods.[3] It is achieved through free energy minimization of intermolecular interactions without external direction. Precise control of self-assembly therefore requires quantitative knowledge of driving molecular forces and the resultant nanoscale structures of the complexation.[4] However, such quantitative knowledge is largely absent as studies of CNM self-assemblies have focused on morphological and functional characterizations. Moreover, in order to gain insight into the driving molecular interactions, systematic studies of the pertinent governing variables are needed, but missing. We have thus investigated the self-assembly of graphene oxide (GO) and a cationic surfactant (cetyltrimethylammonium bromide, CTAB) and quantitatively determined how the structure and interaction of GO:CTAB self-assemblies depend on a variety of solution conditions such as GO size, mixing ratio, salt concentration, and osmotic stress.

GO is readily produced on large scale by oxidizing graphite, which introduces various chemical groups, such as negatively charged COO− and polar −OH groups, along the edges and around defect sites of graphite sheets (i.e., graphene).[5] In contrast with graphene that self-stacks due to strong van der Waals forces, GO is highly soluble in water and can be easily exfoliated as single-atom-layer sheets. It subsequently serves as a very important precursor of graphene that can be obtained by GO reduction, noting that the reduced graphene exhibits inferior, but still competitive, properties compared with pristine graphene.[6]

Facilitated by GO's multifaceted chemistry (i.e., charged, hydrophilic, and hydrophobic groups) as well as its ready reduction into hydrophobic graphene, self-assembly of GO and amphiphilic molecules presents a very versatile platform for designing and creating functional structures through combinations of charge−charge and hydrophobic/stacking interactions. Importantly, amphiphilic molecules such as lipids exhibit wide-ranging self-assembled structures such as liposomes and multilayers in aqueous solution, and they serve as promising scaffolds for directing the assembly of GO, graphene, and other

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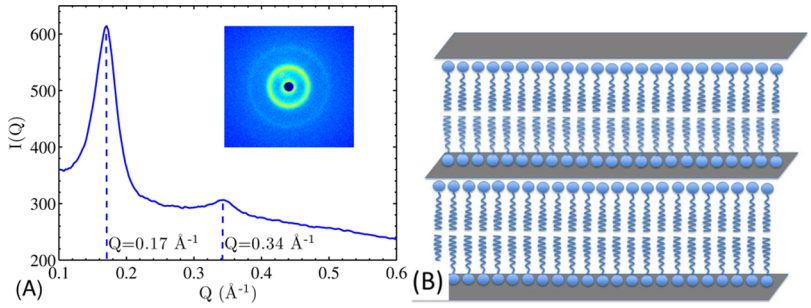


Figure 1. Illustrations of the XRD peaks and multilamellar structure of self-assembled GO:CTAB. (A) Scattering intensity $I(Q)$ of GO:CTAB 1:4 mixture (by weight) as a function of scattering vector $\mathbf{Q} = 4\pi \sin(\theta)/\lambda$, where 2θ is the scattering angle and λ is the X-ray wavelength. The key features are a strong first Bragg peak at $Q = 0.17$ Å^{-1} and a weak but definite second Bragg peak at $Q = 0.34$ Å^{-1}. Inset shows the raw XRD intensity on a 2D image plate detector. (B) Cartoon depiction of the proposed multilamellar structure. GO sheets are shown as planes (gray), and CTAB bilayers are represented with balls and strings.

CNMs.[7−11] For example, an edge-in orientation of GO to positively charged lipid layers was reported at the water−air interface;[12] GO was observed to influence the morphology and stability of the supported lipid bilayers;[13] phospholipid monolayer functionalized graphene was demonstrated to have a high loading capacity of cancer drug DOX; and a novel lipid/graphene conjugate was developed as a fluorescence sensor for detection of phospholipase.[14] In this study, CTAB, a commonly used cationic surfactant, was chosen as a model amphiphile. It is known that CTAB forms stable complexes with GO, and diverse applications have been demonstrated including molecular biosensing,[15] supercapacitor,[16,17] and nanocatalyst.[18] However, despite extensive efforts exploring potential applications of (reduced-)GO:amphiphile self-assemblies, there is a dearth of quantitative studies of the fundamental macromolecular structures and interactions as well as how the self-assembly process is modulated by typical experimental conditions such as salt and composition.

Using X-ray diffraction (XRD) to quantify nanoscale structures, our study reported here systematically probed the driving molecular interactions and resultant structures under a wide variety of experimental conditions. CTAB itself appears to form either membrane-like bilayers or interdigitated monolayers as a result of hydrophobic interactions between their hydrocarbon tails. Interactions between CTAB and GO are shown to be dominated by electrostatics. Multilamellar stacking of CTAB layers and GO sheets persists across wide ranges of GO:CTAB mixing ratios and ionic conditions. While ions are able to intercalate between GO and CTAB surfaces, osmotic stress studies indicate the absence of continuous hydration layers within the multilamellar phase at low salts.

EXPERIMENTAL METHODS

Two batches of GO of different sizes were studied in order to examine their dependence on size. The first was purchased from Sigma (catalog no. 763705, 2 mg/mL dispersion in H_2O, 47 ± 5% carbon in dry mass) and was the main subject of study here unless noted otherwise. The second and smaller GO (carbon mass fraction ∼62%) was synthesized in our laboratory following published methods,[19] and details of synthesis and characterization are described in the Supporting Information. Atomic force microscopy (AFM) was used to check the GO dimensions. The first type from Sigma was observed to mainly comprise GO sheets of sizes >1 μm, and the second type synthesized in-house is predominantly made up of sheets of 200−400 nm ranges. AFM measurements further verified that each type of GO was dispersed as monolayers of ∼12 Å thickness. CTAB was purchased from Sigma (catalog no. 52365) and used as received. Complexation of GO and CTAB was initiated by direct mixing of respective stock solutions in deionized water (DI-H_2O) (both at 2 mg/mL). The mixture was then briefly vortexed before centrifugation at 10000g for 15 min. The precipitate, if present, was loaded onto a microscope slide or X-ray sample cell. Polarization optical microscopy was performed with an Amscope XY-P microscope coupled with a digital camera. XRD measurements were carried out at 20 °C with an in-house instrument which integrates a microfocus fixed Cu-anode source and an image plate detector. Typical X-ray exposure time was 30 min. Data analysis was performed with home-written Matlab codes.

RESULTS AND DISCUSSION

Dispersions of GO and CTAB, when mixed, spontaneously led to macromolecular complexes visible to the eye. When inspected under polarization light microscopy, birefringence was observed, indicating the existence of ordering in the GO:CTAB structure. XRD measurement further showed a pronounced first Bragg peak and a faint second peak for the 1:4 mixture (by weight) of GO:CTAB (Figure 1A). The factor of 2 between the two XRD peak positions indicates multilamellar stacking within the GO:CTAB complex. Given the periodic spacing of ∼38 Å of GO:CTAB assembly and the micron-size scale of planar GO, GO sheets necessarily lie parallel to the stacking plane and are sandwiched between CTAB layers. Taking the GO thickness as ∼10 Å,[20] this leaves ∼28 Å thickness for the CTAB layer which indicates the presence of two CTAB layers between GO sheets,[21] though there exist several compatible structures such as planar bilayers or spherical micelles. Importantly, we also observed an XRD peak at ∼1.51 Å^{-1} and attributed it to the in-plane ordering of the CTAB bilayers with a Bragg spacing of 4.2 Å. A multilamellar phase with alternating CTAB bilayer and GO monolayer as illustrated in Figure 1B was thus revealed. To determine the degree of multilamellar ordering, the first Bragg peak was fitted with a Lorentzian function and gave a full width at half-maximum (fwhm) ∼0.03 Å^{-1}, corresponding to a Scherrer coherent length of 210 Å. The multilayered structure is thus relatively short ranged. One very likely cause is the inhomogeneity of GO in terms of size, shape, surface roughness, and the level of oxidation. The size of GO is a particularly important parameter as it can vary widely with preparation conditions such as initial graphite granularity and

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the degree of sonication. We have thus synthesized a batch of much smaller GO (smGO) in the laboratory and investigated its self-assembly with CTAB in the same way as for purchased GO. In the selected conditions tested, the smGO:CTAB complexes show much broader XRD peaks that can be attributed to the smaller GO dimensions resulting in more "structural defects" in the GO layer, as well as its higher carbon mass ratio indicating less oxygen-containing groups for strong interactions with CTAB. Nonetheless, essentially the same peak positions as from GO:CTAB were observed from smGO:CTAB, indicative of robust multilamellar stacking. In sum, these results establish the formation of multilayered GO:CTAB self-assemblies and quantitate the relevant structural parameters.

As is evident from the ordered multilayer structures, the driving molecular interactions of GO:CTAB self-assembly are hydrophobic forces between CTAB hydrocarbon tails and electrostatic attractions between GO and CTAB head groups. However, as GO surfaces are mixed with negatively charged carboxyl groups, polar hydroxyl groups, and hydrophobic graphitic patches, and interactions between CTAB and GO may involve nonelectrostatic contributions, e.g., hydrophobic interaction between CTAB tails and GO graphitic patches. One pertinent question is whether GO surfaces are completely covered with uniformly charged CTAB bilayers. Here, the measured structural parameters can provide quantitative insight into the composition of the multilayer phase, as illustrated below by assuming the limiting case of full CTAB coverage. The Bragg spacing of 4.2 Å of CTAB layers gives a charge density of 4.9 e/nm^2 and a mass density of 3600 Da/nm^2 for hexagonal packing (5.7 e/nm^2 and 4100 Da/nm^2 for a square lattice, respectively). It is interesting to note that this in-plane Bragg spacing of CTAB is very similar to that of the gel phase of saturation lipid bilayers.[22] As for the GO layer, its surface charge density is not known exactly and varies with the extent of oxidization, which is difficult to control accurately. Nonetheless, as pristine graphene has a mass density of 690 Da/nm^2 from its C−C bond length of 1.42 Å, the overall GO mass density is approximately 1500 Da/nm^2 taking into account of its ∼47% carbon content from Sigma. Therefore, a mass ratio of 1:2.4 for GO:CTAB is expected for a complete coverage of GO surfaces by CTAB bilayers.

We then varied the GO:CTAB solution mixing ratio from 200:1 to 1:200 (by weight) to examine the stoichiometry of their complexation. As GO can be precipitated by centrifugation on its own, pellets were obtained for all mixing ratios. The XRD profiles of selective samples are shown in Figure 2. Among them, GO:CTAB mixing ratios from 10:1 to 1:200 led to samples with pronounced XRD peaks, indicating that CTAB is key to the formation of ordered structures. Note that the GO:CTAB ratios in their complexes are related to but different from the mixing ratios. Figure 3 further shows the Bragg spacing and width of the first XRD peak as functions of CTAB % by weight plotted in Figure 2A. Consistent with a 1:2.4 GO:CTAB mass ratio for stoichiometric GO and CTAB-bilayer complexation, both Bragg spacing and peak width approach asymptotic values above 67% CTAB (i.e., 1:2 ratio). This is further evidenced by appearance of the CTAB in-plane ordering peak at Q = 1.51 Å^{-1} above 1:2 ratio (Figure 2C), supporting rather homogeneous CTAB phases between GO sheets. However, GO surfaces are not homogeneously charged, and full CTAB coverage necessarily leads to overcompensation of GO charges. This is supported by surface ζ-potential measurement of GO:CTAB complexes at varied mixing ratios.

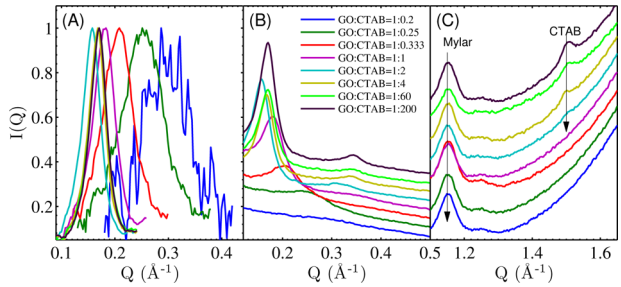


Figure 2. XRD profiles of GO:CTAB self-assemblies at different mixing ratios. The three panels show the different Q regions of each profile that are plotted in the same color. Legends of the colors are given in panel B. All curves are scaled and offset to facilitate visualization. (A) First Bragg peaks after removal of a linear background. (B) Low to medium Q regions illustrate the first and second Bragg peaks. Note that samples with low CTAB fractions give rather broad XRD peaks, and the second Bragg peak only becomes visible from the GO:CTAB 1:2 sample. (C) High Q regions illustrate the CTAB in-plane ordering peak around 1.5 Å^{-1} that becomes visible at high CTAB fractions.

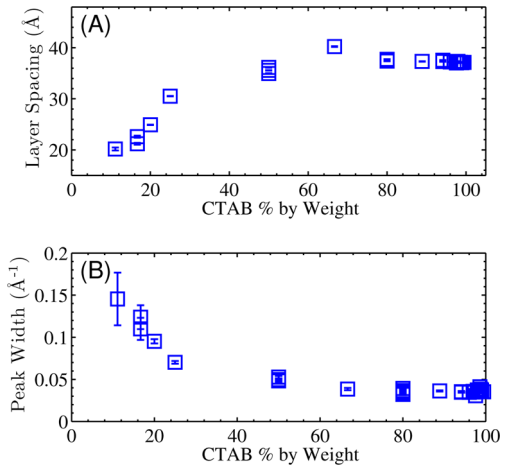


Figure 3. Peak profiles of the first XRD Bragg peak as a function of GO:CTAB mixing ratio. The CTAB % is computed as weight_CTAB/(weight_GO + weight_CTAB); e.g., CTAB 80% refers to 1:4 GO:CTAB mixing ratio. (A) Interlayer spacing of GO:CTAB as a function of the CTAB %. (B) Peak width as a function of the CTAB %.

Charge neutrality occurs at an ∼1:1 ratio, and further addition of CTAB reverses the sign of the ζ-potential. We speculate that CTAB absorption beyond charge neutrality is driven by hydrophobic interactions within the CTAB layers, i.e., between CTAB monomers and absorbed CTAB patches on GO surfaces. Nonetheless, these observations strongly suggest electrostatics as the predominant interaction between GO and CTAB.

At smaller CTAB fractions, the Q = 1.51 Å^{-1} CTAB peak disappears presumably due to the partial removal of CTAB. The loss of CTAB ordering also leads to broader first XRD peaks (Figure 3B) and the disappearance of the second XRD peak (Figure 2B) at CTAB fractions <67%. While a thus weakened GO−CTAB interaction is expected to relax their multilamellar assembly, surprisingly, shorter Bragg spacings are observed at smaller CTAB fractions. Slight decreases in interlayer spacing may be explained by increased tilting of CTAB tails, but spacings as low as ∼20 Å were observed, suggestive of the presence of CTAB monolayers instead of

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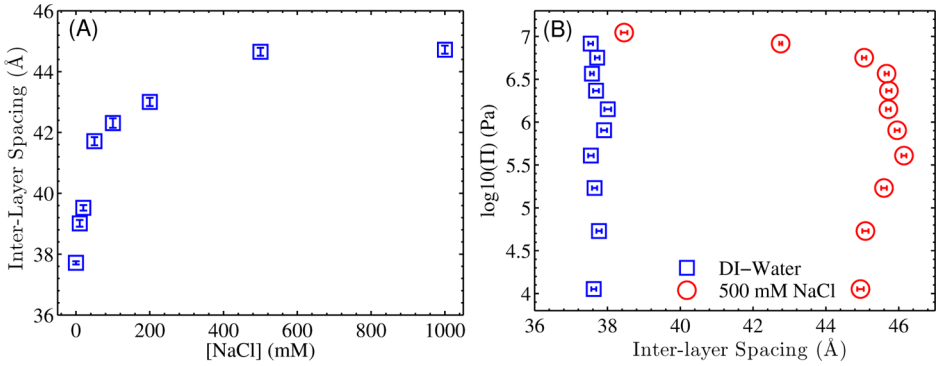


Figure 4. Interlayer spacings of GO:CTAB self-assemblies under varied condtions of salt and osmotic stress. (A) Interlayer spacing is shown as a function of added NaCl. (B) Pressure−spacing relationship is shown for GO:CTAB in DI-H_2O and 500 mM NaCl.

bilayers between GO sheets. We speculate that when the density of CTAB decreases the CTAB bilayer may first become thinner due to tail tilting and eventually collapse into an interdigitated monolayer phase. A further decrease in CTAB density leads to dissolution of GO:CTAB complexes at CTAB fractions below 9% (i.e., 10:1 GO:CTAB mixing ratio). Practically, the studies across a wide range of GO:CTAB mixing ratios delineate conditions for GO:CTAB self-assembly and provide important lattice parameters of the resultant multilamellar structures.

To further probe the dominant electrostatic interactions between GO and CTAB, we then examined the dependence of their self-assembly on monovalent salts (NaCl). The GO:CTAB mixing ratio was kept constant at 1:4 where CTAB is in slight excess, and concentrated salt stock (5 M NaCl) was directly added. Precipitates were observed at all salt concentrations studied (0 to 1000 mM), and all gave the first and second Bragg peaks corresponding to multilamellar structures. As shown in Figure 4, the interlayer spacing monotonically increases with the salt concentration and appears to plateau above 500 mM NaCl. The width of the first Bragg peak only increases slightly. Surprisingly, the CTAB in-plane ordering peak disappears even with the addition of 10 mM NaCl, indicating a more fluidic CTAB bilayer due to ion disturbance. This suggests intercalation of ions between GO and CTAB headgroups which subsequently increases the interlayer spacing. The relatively weak salt effects can be understood in terms of electrostatic competitions between GO and Cl^- and between CTAB and Na^+, whereas GO sheets and CTAB bilayers, both polyvalent, dominate over monovalent ions, respectively. Importantly, this study shows that GO:CTAB complexation is robust in wide ranges of monovalent salts, and that the interlayer spacing can be readily controlled by salt concentration between 38 and 45 Å.

Although electrostatic attraction between GO and CTAB drives their assembly, it is difficult to infer whether their respective hydration shells are retained during complexation. In the resultant multilamellar structure, the presence of significant hydration layers would be important for the stability and diffusivity of the GO and CTAB layers, as well as how it reacts to processes such as freezing and drying. We then applied the osmotic stress method (OSM)[23] to probe the effects of osmotic (water) pressure on GO:CTAB assemblies. In OSM, a neutral polymeric osmolyte (polyethylene glycol of 8000 Da in this study, referred to as PEG8k) was directly added to the supernatant phase of the GO:CTAB mixture, and the sample was rocked to expedite the dissolvation of PEG8k. Due to phase separation, PEG8k in the supernatant phase exerts an osmotic pressure on the GO:CTAB complex, which "squeezes out" the interstitial water molecules and ions that can freely exchange between the two phases. The applied pressure can be varied by the concentration of PEG8k, and the amount of expelled volume can be monitored by the change of interlayer spacing. The concentration of PEG8k was varied in the range of 5% to 50% by weight, corresponding to a pressure range from 10^4 to 10^7 Pa. As shown in Figure 4B, in the pressure range probed ($\sim 10^4$−10^7 Pa), the interlayer spacing remains essentially constant when in DI-H_2O alone, indicating the absence of hydration layers between GO and CTAB. Note that this does not suggest complete absence of water in the GO:CTAB complex, and rather it is suggested that there is not sufficient water to form a continuous layer which the OSM can expel. Note that full dehydration by complete drying in a desiccator led to a decrease of ~5 Å in interlayer spacing, though the cause may be attributed to the change of CTAB layer thickness upon dehydration. Interestingly, at 500 mM NaCl with a larger interlayer spacing (Figure 4B), virtually no change in spacing was shown up to $\sim 3 \times 10^6$ Pa pressure, and then a drop was observed at higher pressures. This observation is consistent with the aforementioned ion intercalation between GO and CTAB layers in high salts and further indicates that high osmotic pressures are required to push ions (and the associated water) out. It is interesting to note that the expelling of ions by OSM is accompanied by the appearance of CTAB in-plane ordering peak around $Q = 1.51$ $Å^{-1}$, in accord with the finding from the salt dependence study that slight ion intercalation disrupts CTAB in-plane ordering.

■ CONCLUSIONS

Using XRD to quantitatively determine the structures of GO:CTAB assemblies, we have probed the roles of several important experimental conditions including GO size, mixing ratio, salt, and osmotic stress (or molecular crowding). In addition to providing quantitative structural parameters, these systematic studies allow us to elucidate the molecular interactions driving GO:CTAB complexation. We found that GO:CTAB assemblies show structural ordering at CTAB fractions >9% by weight. The ordered structures all show multilamellar stacking of GO and CTAB layers, and the interlayer spacing can be varied by mixing ratio, salt, and osmotic stress (when combined with salt) in the broad ranges of 20 and 48 Å. Notably, the CTAB layer takes on a membrane-like bilayered structure with in-plane ordering at high CTAB fractions (>67%), loses in-plane ordering with addition of salt,

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and likely collapses into an interdigitated monolayer structure at low CTAB fractions.

A collection of observations corroborate electrostatic attraction as the driving force of GO:CTAB self-assembly, supported by the sandwiching of anionic GO sheets between cationic CTAB bilayers, GO and CTAB stoichiometry, structural expansion due to ion competition, and the exclusion of hydration layers (in no excess salt). It is worth noting that van der Waals interaction is expected to contribute, but it is difficult to quantify its magnitude. Moreover, hydrophobic interaction may arise between GO graphitic patches and CTAB tails, as previously reported between GO and nonionic Pluronic F127.[12] We attribute the absence of such observation to our focus of GO:CTAB macromolecular assemblies exhibiting structural ordering, as such hydrophobic interactions likely lead to random aggregates. This conclusion is also consistent with previous studies of GO:amphiphile interactions; e.g., Frost at al. reported prevalent electrostatic interactions between GO and lipid headgroups[24] on a supported substrate, and Liu et al. observed electrostatic binding between GO and organic ammonium ions.[25] Notably, ionic interactions have been utilized to effectively disperse GO in organic solvents.[26]

The quantitative studies of GO:CTAB self-assembly reported here shall provide useful guidelines on controlling their structures (e.g., lattice dimensions and ranges) via common solution conditions. Further, richer structure phases may be explored by introducing additional molecular forces. For example, the charge density and apolar nature of GO can be tuned[27] (e.g., by partial oxidation or reduction), and hydrophobic interactions can subsequently regulate the self-assembly.[28] More broadly, future quantitative studies of other CNM self-assemblies (e.g., with polyelectrolytes[29]) are needed to map the comprehensive phase diagrams of attainable nanostructures. Such knowledge will help fulfill the potentials of CNMs in diverse arrays of applications such as in electric nanodevices,[30] sensors,[31,32] composites,[33] and biotechnology.[34−36]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.5b04102.

In-house synthesis of GO; GO characterization with AFM; characterizations of GO:CTAB self-assemblies by polarization optical microscopy and XRD; XRD profiles of GO:CTAB self-assemblies at different mixing ratios; cartoon illustration of the GO:CTAB multilamellar phase with CTAB monolayers; ζ-potential measurement of GO:CTAB mixing series; XRD profiles of GO:CTAB self-assemblies in different salt and PEG8k conditions (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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Supporting Information

Structure and Interaction of Graphene Oxide– Cetyltrimethylammonium Bromide Complexation

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I. In-house synthesis of graphene oxide (GO)

Methods as described by Xu et al. (Aqueous Liquid Crystals of Graphene Oxide. *Acs Nano* **2011,** *5*, 2908-2915) were followed to obtain mono-dispersed GO solutions. In brief, graphite was first oxidized by a mixture of H_2SO_4, $K_2S_2O_8$, and P_2O_5 at 110 ^oC for 6 hours, and a second oxidation step was carried out in H_2SO_4 with gradual addition of $KMnO_4$ and then with H_2O_2. The final mixture was then filtered, rinsed with water, and lyophilized. Note that all excess salts were removed using repeated water rinse and centrifugation before lyophilizing GO. GO solution was obtained by sonicating GO powder in aqueous solution with a Sonics VCX130 unit (6 mm-diameter probe). Typically, 60 mg graphite oxide was sonicated in 30 ml distilled water in ice bath for 2 hours at a power of 30 w. Then the solution was centrifuged at 2000×g for 1h, and very little precipitate was observed. The supernatant was kept for use in the experiments. Success of GO synthesis is indicated by a layer spacing of 9.4(1) Å of dried GO power (Figure S1), and the UV-vis absorption peak at 230 nm of GO (Figure S1 inset).

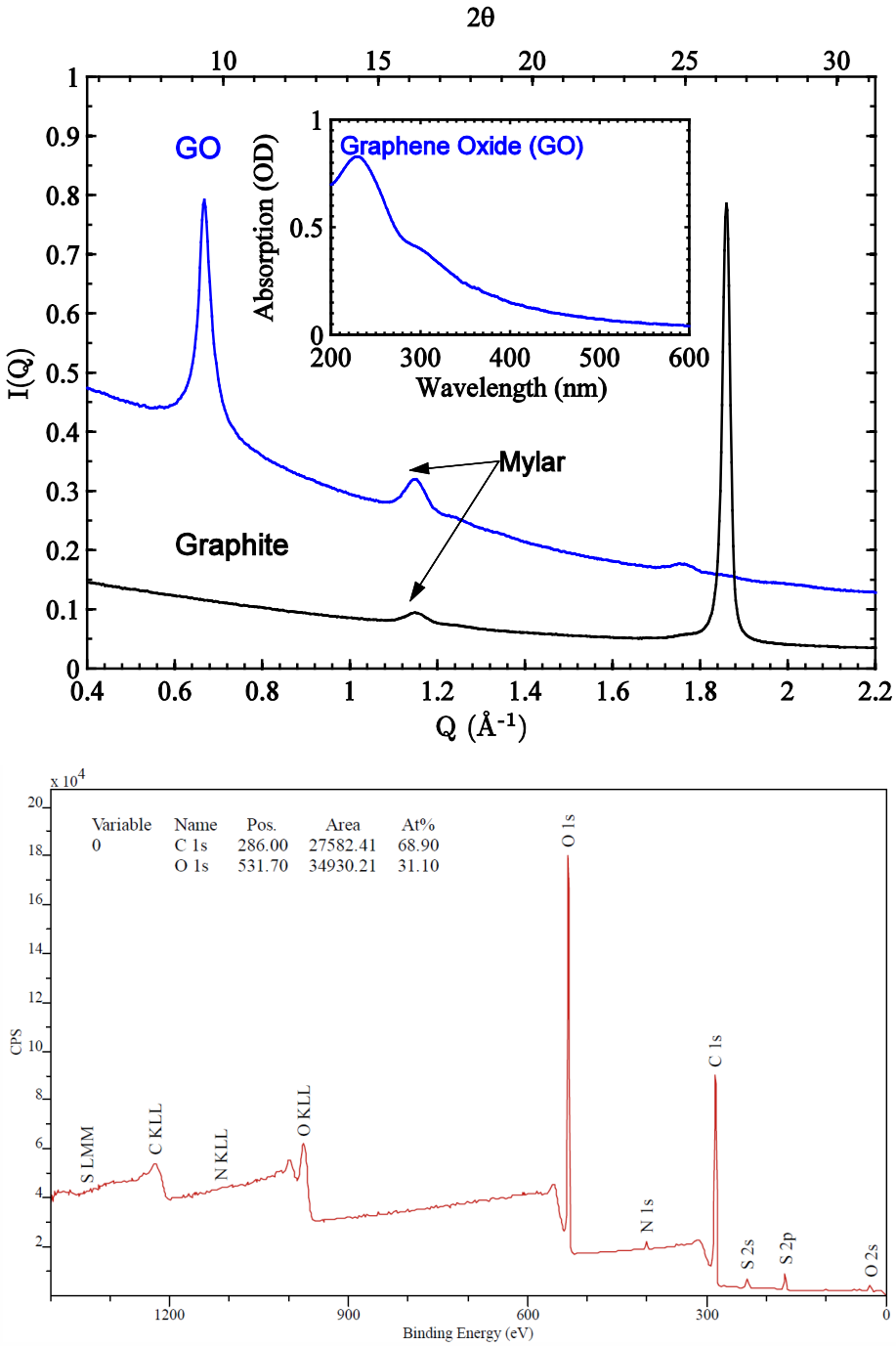


Figure S1. (a) XRD profiles of GO and graphite. The appearance of strong diffraction peak at 2ɵ~9° and disappearance of diffraction peak at 2ɵ~26° demonstrated successful synthesis of GO. Mylar peaks at 2ɵ~16° come from our x-ray sample holder. Inset shows the UV-vis absorption spectrum of GO dispersed in DI-H_2O indicating the GO absorption peak at 230 nm. (b) X-ray photoelectron spectrum of GO synthesized in the lab, giving a carbon mass fraction of 62%.

II. GO Characterization with AFM

Five-µl GO solutions of 0.02 mg/ml were directly pipetted onto freshly cleaved mica surfaces and washed several times with deionized-H_20. The samples were left to dry at 37 °C for >2 hours before imaging. AFM measurements were carried out with an Asylum MFP-3D instrument under tapping mode in air with a TESPA-V2 probe (Bruker^{TM}).

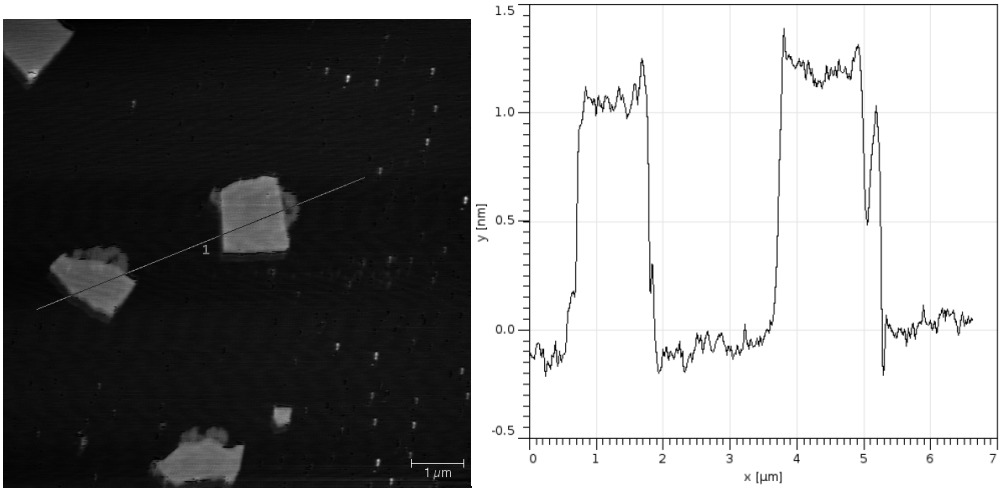


Figure S2. AFM image and line profile of GO purchased from Sigma. The scale bar represents 1 µm. Line profiling shows thicknesses ~12 Å and lateral dimensions >1 µm for most GO sheets.

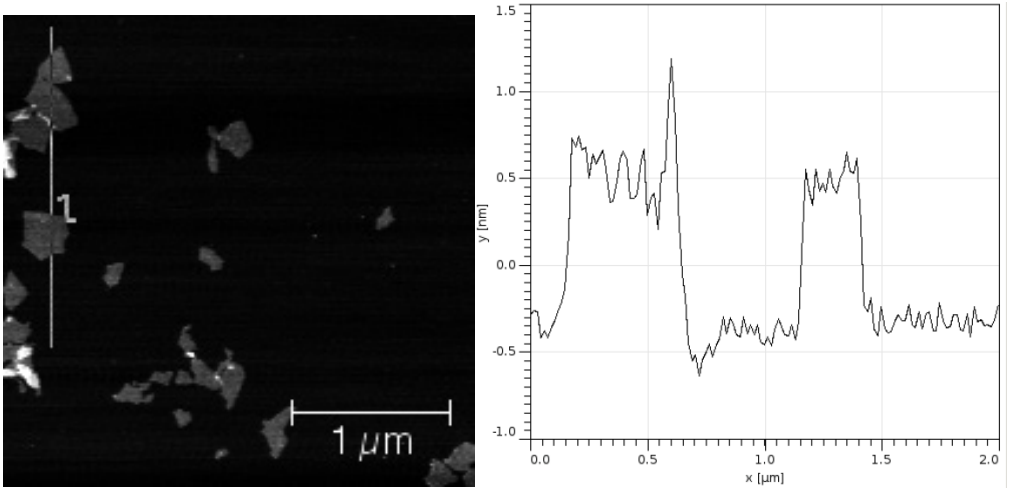


Figure S3. AFM image and line profile of GO synthesized in our lab. The largest GO sheets are smaller than 500 nm and the majority of GO sheets fall in the 200-300 nm range

III. Characterizations of GO:CTAB self-assemblies

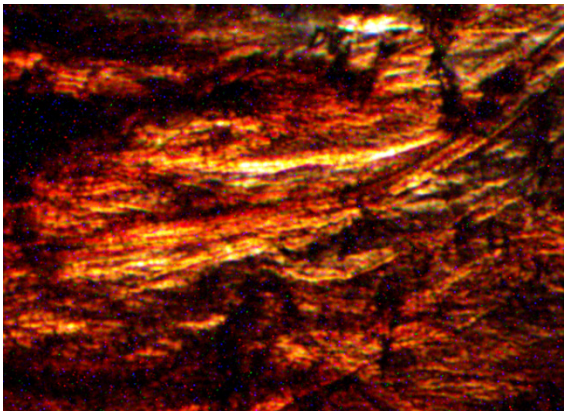


Figure S4. Polarization optical microscopy of GO:CTAB complexes. The full image is 400×300 μm.

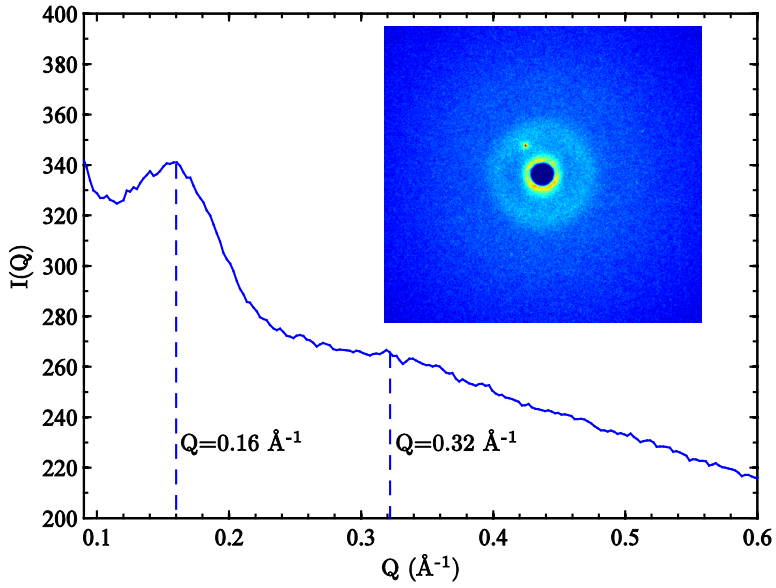


Figure S5. Illustration of the XRD peaks of self-assembled smGO:CTAB at 1:4 weight ratio. The scattering intensity I(Q) is shown as a function of scattering vector $Q=4\pi\sin(\Theta)/\lambda$, where 2Θ is the scattering angle and λ is x-ray wavelength.

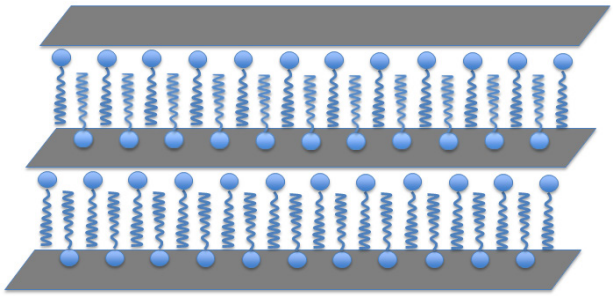


Figure S6. A cartoon illustration of the GO:CTAB multilamellar phase at low CTAB fractions where a monolayer of interdigitated CTAB is suggested.

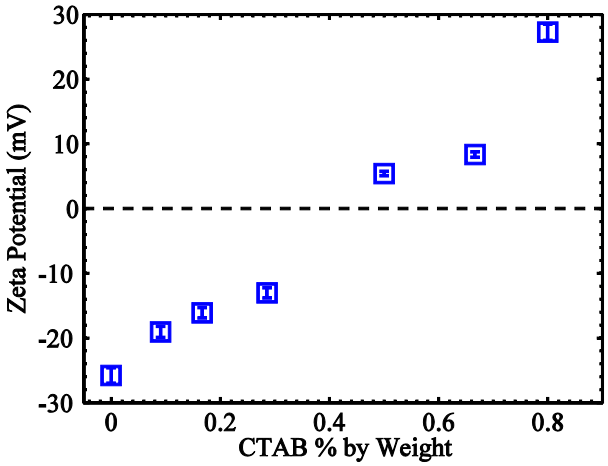


Figure S7. Zeta potential measurements of the GO:CTAB mixing series in de-ionized water. The GO dispersion (from Sigma) was gently centrifuged to remove large sheets and then diluted to 0.08 mg/ml so as to reduce absorption and multiple scattering; concentrated CTAB stock of 1 mg/ml was directly added to the GO dispersion at specific ratio. The Zetasizer Nano ZS90 (Malvern Instrument) was used with a disposable capillary cell starting with ~750 µl of GO solution. At each mixing ratio, three runs were collected and their values and respective errors were averaged. It is worth noting that the reported error for each run is usually smaller than the variation between runs and we attribute to the likely association of GO and CTABs to form micro-aggregates. Overall scattering counts also shows steady decrease with the addition of CTAB, thus the results of zeta potential measurements should be considered as being qualitative.

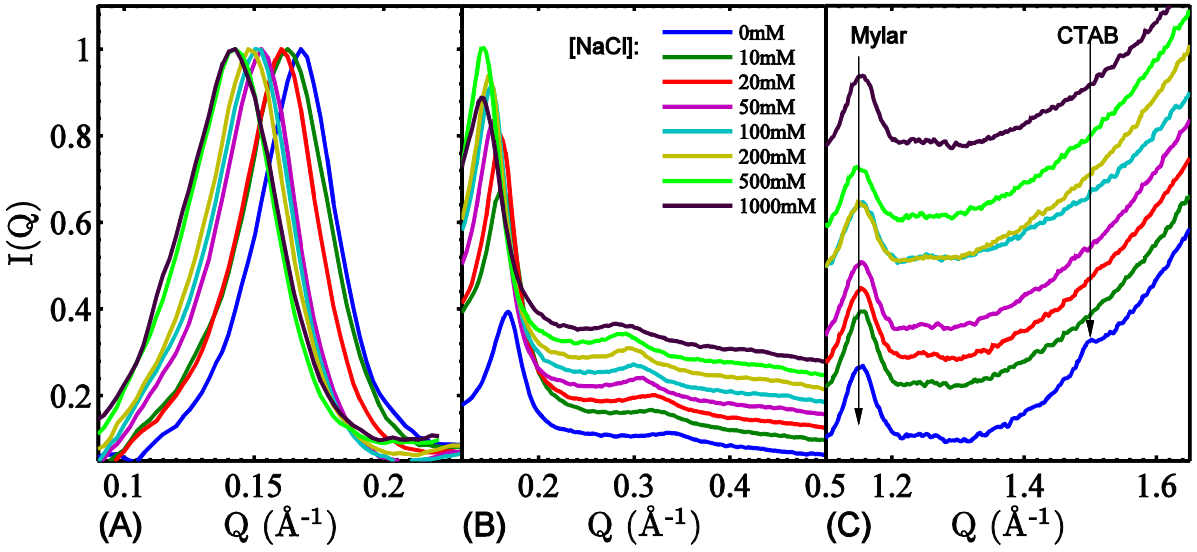


Figure S8. XRD profiles of GO:CTAB self-assemblies in different salts. The three panels show the different Q regions of each profile that is plotted in the same color. Descriptions of each panel follow that as in Figure S6.

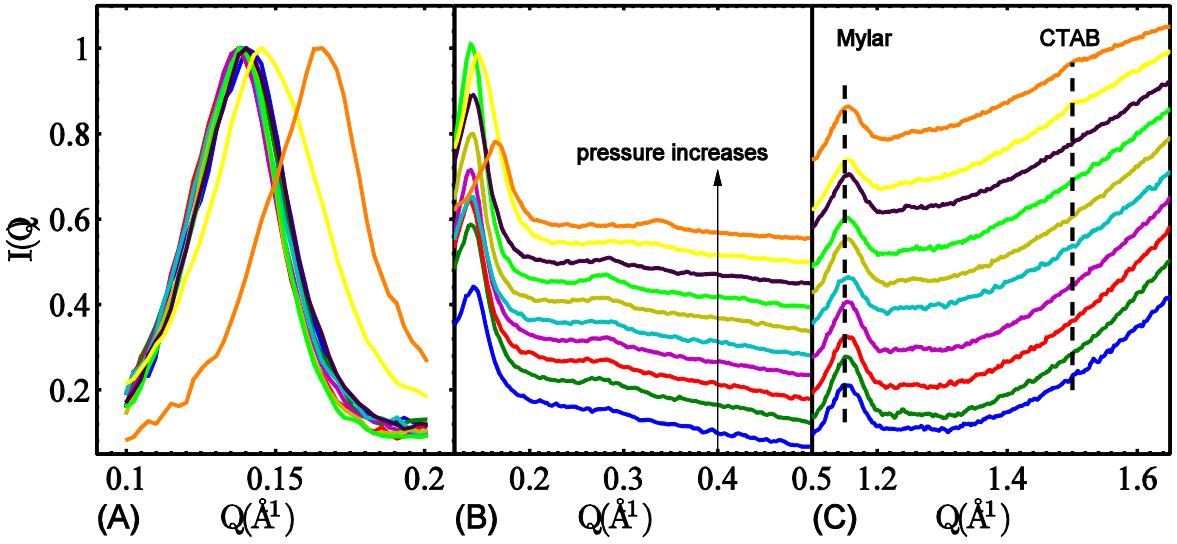


Figure S9. XRD profiles of GO:CTAB self-assemblies in 500 mM NaCl under different osmotic pressures. Descriptions of each panel follow that as in Figure S6. Multilamellar structures are observed under all conditions as evidenced from the 2nd Bragg peaks in panel B. The CTAB in-plane peak starts to emerge, though very faint, at highest pressures probed.